

Business Research Project – Economics



Beyond the Resource Curse: Modeling the Structural Trade-offs of Mineral-Led Development in Pakistan

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Abstract

While Reko-Diq has succeeded in drawing global attention, its empirical assessment and significance towards Pakistan needs rigorous study. The research investigates the significance of Reko-Diq project which is widely termed as the game-changer for Pakistan's economy, to test macro-economic stability *with* and *without* Reko-Diq, aiding in constructing comparative assessment of project's impact on Gross Domestic Product (GDP) and Current Account (CA).

To empirically investigate the structural determinants of Pakistan's economic growth and external balance, this study employs the Autoregressive Distributed Lag (ARDL) Bounds Testing approach using annual time-series data from 1998 to 2024. Following the confirmation of long-run cointegration of variables via the F-Bounds test, the study utilized the models and their historical coefficients to generate dynamic ex-ante forecasts up to the year 2065, simulating the structural impact of the Reko Diq project under distinct economic scenarios.

The research illustrates that Pakistan's nominal GDP grows better without Reko-Diq and in contrast improvement in the Current Account position of Pakistan by the end of the project majorly due to higher mineral export.

These findings are essential as they help in understanding that even though Reko-Diq strengthens external account, it must still not be considered as a sole rectifier of the macro-economic variables.

Chapter 1

Introduction:

The Global energy systems are rapidly moving towards renewable energy and mineral intensive advancement which requires significant supply of strategic minerals such as Gold and Copper. Less Developed Nations (LDCs) with rich supply of strategic minerals such as Pakistan are presented with an opportunity to develop themselves with reserves projects such as Reko-Diq. The project which is often considered as a transformative project for Pakistan due to its potential and massive scale might still pose macro-economic and structural challenges for Pakistan due to state's prevailing reality. Such projects are widely considered as a solution for macroeconomic hurdles, however relationship between resource abundance and growth rate is not always linear (Resource Curse).

As discussed above, Reko Diq potential is widely studied in political, legal, qualitative, and even numeric dimensions. However, Reko-Diq's significance for Pakistan based on state's historical and empirical stature is needed to be studied. In this study, we will understand how Pakistan behaves to increased influx of mineral resources and government expenditure and whether inflating these variables supports growth or causes structural imbalances, probably crowding out?

The research aims to fill this gap with the help of dual-model approach that answers two questions, 1. Is Reko-Diq significant for Pakistan's Gross Domestic Product? 2. Is Reko-Diq capable of alleviating Pakistan's chronic current account imbalances?

By analyzing historical time-series data from 1998 to 2024, the research establishes long-run cointegration among key variables while utilizing the Autoregressive Distributed Lag (ARDL) Bounds Testing approach. These historical coefficients are then used to generate dynamic *ex-ante* forecasts up to the year 2065, simulating the Pakistani economy under two distinct scenarios: a baseline 'without-project' trajectory and a 'with-project' scenario incorporating Reko Diq's projected flows.

The following research suggests a trade-off between two variables of Pakistan, a remedy for Current account while dampening Gross Domestic Production growth rate. Understanding these dynamics is essential for designing fiscal policies that can harness the external benefits of Reko Diq while mitigating its internal structural costs.

Chapter 2

Literature Review:

The paradox of plenty, where natural resource abundance correlates with poor economic performance remains one of the most contested debates in development economics. While foundational literature posits an inherent negative relationship between resource intensity and growth due to Dutch Disease and rent-seeking. Authors from around the World terms it as 'Resource Curse' as they argue that resource abundance leads to poor governance and increased corruption hence lowering growth. While natural resources offer economic potential, they often become the underlying cause of strategic mismanagement and institutional decay where countries dependent on natural resources fail to develop tax bases or strategic institutions. It also states that such resources provide incentives to foreigners/third parties to perpetuate conflicts in such regions as they aim to gain access to such resources. Humphreys (2005) termed this as 'weak state mechanisms' and 'Greedy Outsiders Mechanism'.

Furthermore, resource abundance increases the likelihood that Less Developed Countries (LDCs) evolve into "predatory states" rather than working on an industrial model which is competitive and can aid in development (Auty 2001). He further posits that such nations tend to rely on natural resources rather than structural reforms and measures hence causing poor governance and economic infrastructure. Auty (2007) critically explains the consumption pattern of resource abundant nation, it ites that these states consider this influx of money as permanent causing reduced savings. It is to note that the stream of income from such resources are termed as 'volatile'.

Stijn's (2000) highlighted that nations who adapt to defined property right system tend to succeed in development through resources such as Developed Countries (DCs) while arguing that Dutch disease symptoms are present in oil and gas abundant countries. Corden and Neary (1982) contributed with their 'Booming Sector' model while aligning with 'Dutch disease' which explained how resource boom leads to appreciation of currency and hence lowering competitiveness of other export commodities. Gylfason (1999) expanded on this with a two-sector model of tradable and non-tradable goods followed by Humphreys (2005) who explained the foundational 'Greedy Outsiders' hypothesis while analyzing the role of third parties in instigating conflicts to secure resource access. Wizarat (2013; 2014) expands this framework by stating that states dependent on natural resources or agricultural commodities rather than a broad taxation base often have weak state structures, making them vulnerable to external control. Wizarat further deepens Humphreys' analysis by linking resource extraction to the "industrial-military complex", arguing that conflicts which are initiated by greedy outsiders generate a secondary revenue stream for developed nations through the arms trade, hence creating a self-reinforcing cycle of instability.

While the resource curse literature has extensively provided the adverse growth, governance, and conflict associated with mineral dependence, this perspective is not universally accepted. It is

also discussed that resources are not cursed as their result depends on the effectiveness of governance, institution and corruption. Mehlum et al. (2006) cited this as ‘Paradox of Plenty’ while distinguishing between ‘grabber friendly’ and ‘producer friendly’ nations. Bulte et al. (2005) discussed the point-resources while explaining that areas of resources are small and hence can be easily controlled by small groups hence causing skewed income distribution.

In context of Pakistan, mineral resources are significantly untapped, meaning that the state has yet to practice the full potential of its geography for its geo-strategic, political and economic benefits. As the world shifts towards renewable and sustainable energy practices, LDCs holding copper, and rare earth elements have new geopolitical and economic leverage. Furthermore, mineral availability is intrinsically tied to defense capabilities and the expanding arms race, which Wizarat (2013) classifies as ‘military might’ while further elaborating it as ‘brute force and military strength’, contained by well-developed and resource rich nations. Other than this, mining royalties and taxes help LDCs fund their health and education as well as development of infrastructure which benefits other sectors and industries as stated by ESIA (2024-25) which terms it as “social uplift”.

With the given literature and framework of previous research works, we can analyze Pakistan’s massive Copper and Gold reserves of Reko Diq; while understanding the challenges and opportunities state might encounter in its attempt to hold strategic and economic significance of this project. Arif Habib Limited (2022) analyses Reko Diq project as a massive capital structure and world-class mineral deposits with a total capital expenditure of USD 7 Billion. Production forecasts are equally robust; annual copper output is targeted at 650–700 million lbs. for the first decade, rising to 800–850 million lbs. post-expansion. Simultaneously, gold production is expected to commence at 300k–350k oz annually, eventually increasing to 450k–500k oz after the second phase is completed.

While the previous research papers address the resource curse and Dutch disease alongside governance, institutional effectiveness, and geology of the project there is a lack of **data-driven assessments** that model whether Pakistan’s current economic indicators (GDP, FDI, institutional quality scores) are sufficient to absorb the Reko Diq windfall without triggering the conflict/inflation transmission mechanisms identified by Wizarat and other authors. Hence for Pakistan to fully practice Reko-Diq’s strategic strength, state capacity and security strength is essential to deter "Greedy Outsiders" from turning Baluchistan into a conflict zone to extract cheap copper.

Chapter 3

Methodology:

This study adopts a quantitative and descriptive-analytical research design. The research relies on secondary macroeconomic and institutional data to assess variable's historical relationships leading to simulating long-term economic trajectories.

The data is observed from 1998 till 2024 with annual frequency to conduct econometric estimation while the period 2025 till 2064 is used for scenario-based simulation. The variable selection is annual while ensuring sufficient observations are used.

Macro-economic variables of Tax, Foreign Direct Investment (FDI), and Government Expenditure figures sets were extracted from *World bank* database, Worker's Remittances from *trading-economics*, REER Index from *State bank of Pakistan*, Minerals Net Export from *The Observatory of Economic Complexity* and Labour force from *Pakistan Bureau of Statistics*. These sources were selected due to their coverage, consistency and accuracy for empirical analysis.

In addition to historical macroeconomic data, project-specific forecasted figures including net cash-flows, revenue streams, tax and royalty revenue and mineral export volume is derived from Barrick Gold technical feasibility report. In this report, it is important to know that figures incorporated are treated as exogenous inputs for analysis purpose only. These figures do not influence the estimation output however are introduced during simulation phase to assess the potential structural impact of the project.

The forecasted figures were later incorporated in the baseline models which led to the formulation of modified models, namely "GDP_RD" and "CAGDP_RD". Thus, allowing a direct comparison between baseline scenario and modified scenario that takes into account the project. The projection must be interpreted as conditional scenario-based simulations rather than forecasts as they are dependent on ceteris paribus behavior and estimated relationship of the variables over the projection horizon of 40 years.

3.1. Variable Forecasting :

Variable	Growth Rates			Description
	Phase 1	Phase 2	Phase 3	
Mineral Export Deficit (MXD)	3%	2%	0.1%	Forecasted by first inflating import values using the recent trend for the initial years. Export values were also increased at a lower rate than imports. This approach captures the expected long-term expansion of both import and export flows while preserving the structural gap between them. Most importantly, it incorporates expected export values of both Copper and Gold from the project for <i>with Reko-Diq scenario</i> .
Remittances	9%	6%	3%	To forecast remittances over the 2025–2065 horizon, this study employs a three-phase 'Step-Down' growth model to prevent exponential outliers. This tapered approach ensures projections remain structurally realistic relative to GDP. Remittances remain same for <i>without</i> and <i>with Reko-Diq scenario</i> .

Primary Income Deficit (PI)	6.8%	5%	4% - 2%	Primary income is forecasted by applying the average growth rate of the existing negative balance in the initial years and then shifting to a gradually declining path. Since primary income values are negative, lower growth over time reflects an improvement in the balance, consistent with expectations of structural reforms and the reduced profit outflows associated with multinational exits. The projection also includes the scheduled payments made to Barrick Gold by RMDC for <i>with Reko-Diq scenario</i> .
USD foreign exchange rate to PKR (FX)	6%	4%	4%	Forward exchange rate path was constructed based on Pakistan-specific inflation expectations. PKR/USD exchange rates were projected by applying an assumed annual PKR inflation rate. This methodology directly accounts for PKR's expected loss of purchasing power, consistent with the conversion of real USD values (which hold constant purchasing power) into nominal local currency terms. Under <i>with Reko-Diq scenario</i> , currency was stabilized due to increased forex inflows in the economy.
Gross Fixed Capital Formulation (GFCF)	-0.23%	0.03%	0.02%	Forecasted based on historical average which was increased at a greater rate under <i>with Reko-Diq scenario</i> as compared to <i>without Reko-Diq</i> .
Government Expenditure (GE)	8%	5%	2%	Government expenditure is projected using a gradually moderating growth path that reflects both recent fiscal behavior and the expected stabilization of spending over the long term. The forecast uses a rate close to the historical average for the initial years, capturing current expenditure patterns as well as the higher government outlays associated with the early phase of the Reko Diq project for <i>with Reko-Diq scenario</i> and steadier rates for <i>without Reko-Diq project</i> .
Labour Force	1.5%	1.5%	1%	The labor force forecast is based on historical averages for recent years, adjusted for anticipated employment generation from the Reko-Diq project. Initial years reflect a spike in employment during the construction phase, followed by steady growth, and a tapering in later years.

Table 1

Note: Growth rates are chosen conservatively to avoid explosive trajectories

3.2. Model and Framework:

The two models included in the research defines functional relationship between variables:

Current Account Model:

$$CAGDP_t = \beta_0 + \beta_1 \Delta FX_t + \beta_2 \ln(MXD_t) + \beta_3 \ln(PID_t) + \beta_4 \ln(remittances_t) + \epsilon_t$$

Investigates impact of Exchange rate (ER), Mineral Export Deficit (MXD), Primary Income Deficit (PID) and Workers Remittances with and without Reko-Diq on the Current Account to GDP ratio of Pakistan.

Gross Domestic Product Model:

$$\ln(GDP_t) = \beta_0 + \beta_1 \ln(GFCF_t) + \beta_2 \ln(lbrforce_t) + \beta_3 \ln(GE_t) + \epsilon_t$$

Investigates impact of Government expenditure (GE), Labour force (LBRFORCE) and Gross Fixed Capital Formation (GFCF) with and without Reko-Diq on the GDP of Pakistan.

To comprehend the specific impact of the Reko Diq, the study constructs two distinct economic scenarios for the period 2025–2065:

3.2.1 Scenario A: The Baseline (Counterfactual) Scenario

This scenario represents the historical trends observed from 1998 to 2024, continued without the intervention of the Reko Diq project.

Independent variables namely Government Expenditure ($\beta \ln(GE)$) and Mineral Export Deficit ($\beta \ln(MXD)$) are extrapolated with their historical average growth rates and a conservative approach for the later decades.

This serves as the control group, explaining what Pakistan's GDP and Current Account would look like in the absence of the Reko-Diq project.

3.2.2 Scenario B: The "With-Project" Scenario

This scenario takes into consideration the exogenous financial shocks associated with the Reko Diq project. The independent variables are manually adjusted based on feasibility data and official projections from Barrick Gold feasibility reports. Such as Labour force, mineral export is increased whereas primary income is reduced hence increasing deficit size.

- Shock 1 (Trade Balance): The Mineral Export Deficit ($\beta \ln(MXD)$) and ($\beta \ln(FX)$) are adjusted to reflect the inflow of copper and gold export revenues, reducing the trade deficit and strengthening Pakistan Rupee (PKR).
- Shock 2 (Outflows): Primary Income Deficit ($\beta \ln(PID)$) is adjusted to account for profit repatriation and dividend outflows to foreign stakeholders.
- Shock 3 (Fiscal Expansion): Government Expenditure ($\beta \ln(GE)$) is increased to reflect the state's expenditure in setting up the project.
- Shock 4 (Employment generation): Labour force ($\beta \ln(lbrforce)$) is increased to incorporate job creation due to the project.

The simulation conducted in the research is 40 years long, which is the expected life of the project. The research aims to construct thorough comparison between **Scenario A** and **Scenario B** aiding to formulation of net marginal impact on the economy of Pakistan.

3.3. Estimation Strategy

Autoregressive Distributed Lag (ARDL) Bounds Testing approach was used as one of the estimation techniques as it works well with small sample sizes followed by Augmented Dickey-Fuller (ADF) tests to ensure no variables is $I(2)$. F-Statistics were used to ensure long-run relationship whereas long-run coefficients were used to estimate the magnitude of impact. Serial Correlation, Heteroskedasticity and stability were conducted to ensure model's efficiency, followed by ARDL framework to perform dynamic forecasting and scenario analysis. This phase aims to simulate the structural impact of the Reko Diq project on Pakistan's macroeconomic indicators (GDP and CA) over the project's estimated lifecycle.

3.4. Limitations and Robustness of the Forecasts

To address the limitation of a forecasting horizon (2025–2065) exceeding the estimation sample (1999–2025), this study utilizes the ARDL bounds testing approach, known for its robustness and consistency in finite samples. The forecast employs an ex-ante structural simulation rather than simple trend extrapolation, relying on the validity of the underlying economic relationships. Crucially, the model's robustness is empirically verified via CUSUM diagnostic tests (**Figures 5 and 6, see Appendix C**), which confirm that the structural parameters, specifically the elasticities of Labor, Capital, and Mineral Exports remain strictly within the 5% critical bounds. This stability validates the use of historical coefficients for long-term simulation, provided the structural "laws" of the economy remain consistent under *ceteris paribus* conditions.

Consequently, the results should be interpreted not as unconditional predictions, but as conditional scenarios dependent on the stability of these proven structural relationships under *ceteris paribus* conditions.

Chapter 4

Empirical Analysis:

4.1. Unit Root Test:

Prior to estimation, Augmented Dickey-Fuller (ADF) tests were conducted to assess the stationarity of the variables while the results confirmed **variable integration at mixed orders of $I(0)$ and $I(1)$** .

4.2. Model Specification:

The study employs the Autoregressive Distributed Lag (ARDL) approach to examine the dynamic relationships between the variables. The optimal lag length for both models was selected automatically based on the Akaike Information Criterion (AIC) which selected an ARDL (1, 0, 0, 0) structure for GDP model and ARDL (1, 0, 0, 0, 0) structure for CAGDP model, indicating one lag of the dependent variables (GDP) and (CAGDP) while the independent variables influence the dependent variable contemporaneously (with zero lags).

Goodness of Fit and Diagnostics of both the models are shown in Table 1Table 2.

Goodness of Fit and Diagnostics

Model	R-Squared	F-Statistics	F-Stat(Prob)	Durbin-Watson
GDP	0.98	261.6	0.0000	2.14
CAGDP	0.75	13.02	0.0008	2.09

Table 2

4.3. ARDL Bound Test:

To determine and ascertain long-run relationship between GDP and Current Account to GDP ratio of Pakistan with their structural determinants, ARDL long-run form and bound tests were conducted.

ARDL Long-run Form and Bound test

Model	F-Statistic	Upper Critical Bound ($I(1)$)
GDP	9.18	4.66
CAGDP	6.55	4.37

Table 3

The result above shows strong statistical evidence of cointegration between dependent variables and their determinants, hence we ascertain **long-run relation of GDP and CAGDP models with their variables, rejecting null hypothesis of No Cointegration (no long-run relationship exists amongst variables).**

4.4. Long-run Coefficients:

Long-run coefficients of the **CAGDP model** (Table 4) show a positive relationship of Workers Remittances having a coefficient of 0.0696 with the Current Account to GDP ratio of Pakistan ascertaining that remittances inflow inflates Current Account balance.

On the contrary, Mineral Export Deficit, Primary income deficit and Exchange rate increment creates hurdles for the nation's CAGDP model with coefficient of -0.0162, -0.0849 and -0.0014 respectively. Hence, we are able to understand that worsening mineral export and primary income balance derails current account balance of Pakistan which aligns with the economic theory of Mercantilism and structuralist perspectives on trade balance. Similarly, the Exchange rate coefficient depicts classical Pakistan's situation where currency depreciation negatively affects the Current Account position. This is likely due to the inelastic nature of imports where a weaker Rupee significantly inflates the import bill, outweighing any benefits from cheaper exports.

Long-run coefficients of the **GDP model** (Table 5) show a positive relationship of Labour Force and Gross Capital Formation (GFCF) having a coefficient of 4.0287 and 0.7399 respectively with the GDP of Pakistan ascertaining that increase in employment significantly and capital formation marginally contributes in GDP growth, aligning with economic assumptions and ascertaining Pakistan as Labour intensive economy.

On the contrary, Government Expenditure with coefficient -0.3438 negatively affects GDP growth. This is primarily the case of 'crowding-out' in Pakistan as Government tends to increase expenditure through borrowing from commercial banks leading to reduced funding opportunities

for private sector and at a higher interest rate. The data and results therefore indicate Pakistan's inefficiency with rising government expenditure.

ARDL Long-run coefficients of the CAGDP model

Variables	Coefficient	Std. Error	T-statistics	Prob
ln(Mineral Export Deficit)	-0.0178***	0.006	-2.993	0.007
ln(Remittances)	0.0676***	0.024	3.017	0.007
Δ Exchange rate	-0.0014**	0.005	-2.721	0.014
ln (Primary Income Deficit)	-0.0798**	0.035	-2.498	0.026
Constant	0.7299*	0.402	2.067	0.064

Table 4

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, Author's calculation, Δ denotes the first difference operator; and $\ln(\cdot)$ denotes natural logarithm.

ARDL Long-run coefficients of the GDP model

Variables	Coefficient	Std. Error	T-statistics	Prob
ln(Government Exp)	-0.3438***	0.006	-2.993	0.008
ln(Labour Force)	4.0287***	0.024	3.017	0.000
ln (GFCF)	0.7399***	0.035	-2.498	0.007
Constant	-53.470***	0.402	2.067	0.000

Table 5

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, Author's calculation, Δ denotes the first difference operator; and $\ln(\cdot)$ denotes natural logarithm.

4.5. Short-run Coefficients (Error Correction Terms):

Error Correction Terms indicates rapid speed of adjustment from disequilibrium to equilibrium of GDP, approximately 83% of disequilibrium caused in previous year converges to long-run equilibrium in the current year. Apart from this, Current Account has a moderate speed of adjustment, approximately 45% of disequilibrium is rectified in the current year which means it takes over two years to fully adjust from disequilibrium.

Error Correction Form Table

Variable	Error Correction Terms	Significance
GDP	-0.83	Highly Significant
CA/GDP	-0.45	Highly Significant

Table 6

4.6. Research Findings:

The research concludes with comprehensive results, taking into consideration forecasted values of the macro-economic variables and testing them under two scenarios, 1. *With Reko-Diq* and 2. *Without Reko-Diq*.

1. Pakistan with Reko-Diq

This scenario shows the future forecasted figures of Pakistan's GDP and Current Account over the life of the project, incorporating the lifecycle financial flows of the mine. The figures depict 5.2% average annual growth rate of GDP over the course of 40 years whereas Pakistan's Current Account position illustrates widening deficit in the phase 1, followed by recovery in the phase 2 and eventually exponential growth in the surplus over the remaining life of mine (LOM) from 2050-2064. The current account position illustrates classical case of J-Curve which is common for a project like Reko-Diq as initial years are most likely to be hefty on the trade balance of Pakistan due to machinery and technical import. The curve turns sharply from 2050 onwards, reflecting a period where inflows from copper and gold exports takes over hence resulting in exponential surplus growth till the end of the life of mine (LOM). The trend line of absolute GDP and Current Account to GDP ratio of Pakistan under Reko-Diq umbrella is shown in *Figure 1* and *Figure 2*, respectively.

2. Pakistan without Reko-Diq

Conversely, this scenario takes into consideration the baseline trajectory of Pakistan devoid of Reko-Diq project intervention. The research depicts 5.3% average annual growth rate of GDP, marginally higher than Reko-Diq scenario. The 0.1% differential is not a loss of economic output, but rather a result of the Exchange Rate Stabilization effect. In the absence of Reko-Diq, currency depreciation fuels import-cost inflation, artificially inflating the nominal GDP figures which has been the case of import dependent nation like Pakistan. Conversely, the resource inflows from Reko-Diq stabilize the currency, dampening domestic price levels.

Pakistan's Current account position shows significant structural deficit on an increasing trend. This exhibits that without export diversification provided by the mining sector; Pakistan's current account will not be able to withstand import pressure and exchange rate devaluation in the long-run. Furthermore, this indicates that the state would require consistent external financing to bridge the gap of its current account position. The trend line of absolute GDP and Current Account to GDP ratio of Pakistan without the project is shown in *Figure 1* and *Figure 2*, respectively.

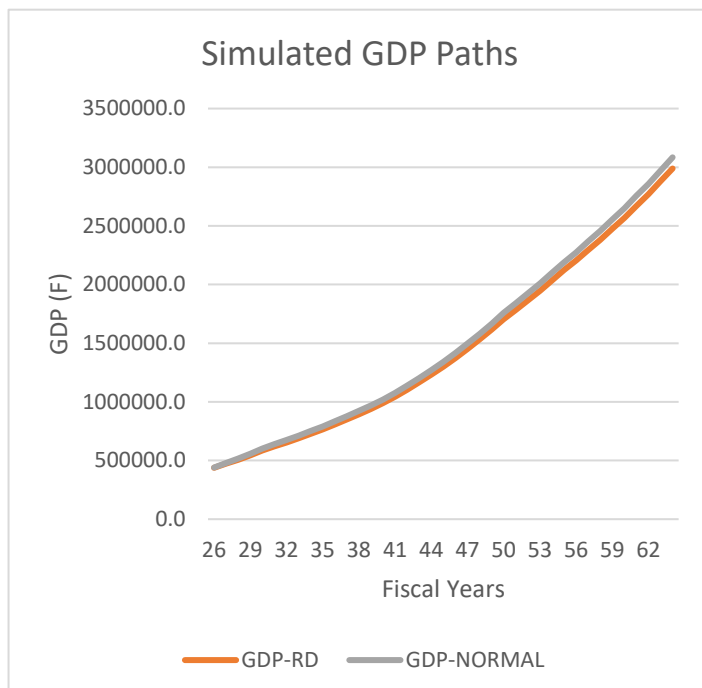


Figure 1

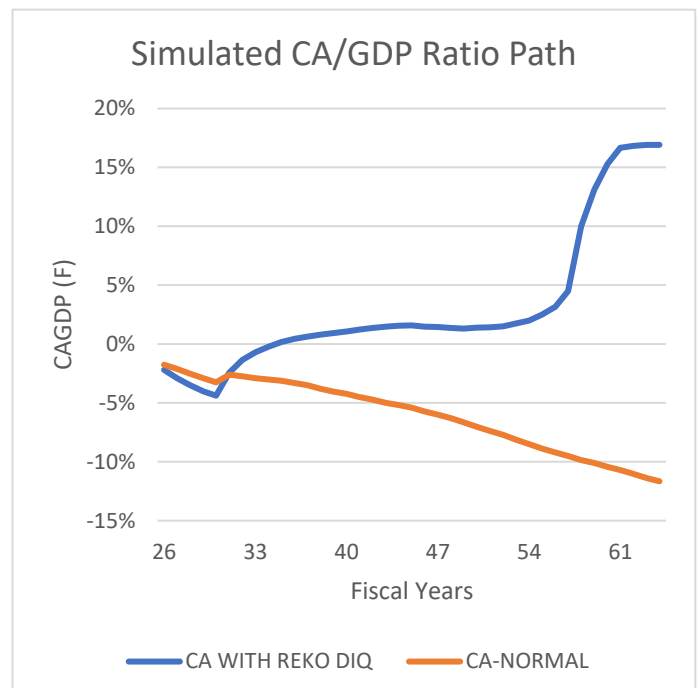


Figure 2

Chapter 5

Research Conclusion:

5.1. Interpretation:

The study manages to quantify the impact of Reko-Diq on macro-economic variables of Pakistan with the help of counter-factual scenario. The empirical findings of the research reveal that while Reko-Diq does not contribute in accelerating nominal GDP growth, it serves as a critical stabilizer and generator of revenue stream for Pakistan's external balance.

Based on the ARDL estimation and forecasting, Pakistan's current account worsens without the project and gains marginally for its nominal GDP. Research has further illustrated J-Curve formation for the current account which depicts trade deficit strain in the initial years followed by stream of surplus, hence fundamentally securing country's external trade position.

Notably, based on the historical data estimation output we were able to validate the "crowding-out" phenomenon; a rise in government expenditure hindering GDP growth while identifying Labour force as a significant driver of the economy. In this research, we can also establish the presence of currency depreciation as a driver of current account deficit in the long run.

5.2. Policy Recommendation:

Policy Recommendation	Reasoning
Currency Stabilization	Currency depreciation creates hurdles for the Current account position of Pakistan, though marginally. Policy makers must ensure currency stabilization policies are used to ensure trade balance improves.
National Resource Fund	Influx of massive forex currency due to current account surplus post 2050 will lead to significant appreciation of currency while the objective is stabilization, hence formulation of resource fund entity will allow ample investment in offshores or long-term domestic capital formulation for the government.
Expenditure Dampening	Government expenditure such as subsidies, pensions and salaries dampens nominal GDP growth of the country, privatization of non-strategic entities must be prioritized to curtail additional burden on the government.
Labour Inclusion	Training and development of labour is essential to achieve robust nominal GDP growth.
Capital Formation	Policy makers must prioritize capital formation with Public-Private partnership, China-Pakistan Economic Corridor (CPEC) Phase 2 shall be aimed at generation of investment and capital gain across the country.
Institutional Development	As Wizarat (2006) and Auty (2001) emphasized on institutional and governance quality to fully benefit from Resource abundance, it is essential for policy makers to curtail involvement of “Greedy Outsiders” with the help of transparent and equitable processes and contracts.

Table 7

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Appendix A

Dependent Variable: CAGDP
Method: ARDL
Date: 12/11/25 Time: 17:22
Sample (adjusted): 1999 2025
Included observations: 27 after adjustments
Maximum dependent lags: 1 (Automatic selection)
Model selection method: Akaike info criterion (AIC)
Dynamic regressors (0 lag, automatic): LOG(REMITTANCES) D(FX)
LOG(MXD) LOG(PID)
Fixed regressors: C

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
CAGDP(-1)	0.550832	0.122842	4.484073	0.0002
LOG(REMITTANCES)	0.031283	0.008850	3.534750	0.0020
D(FX)	-0.000669	0.000168	-3.970425	0.0007
LOG(MXD)	-0.007293	0.001712	-4.261332	0.0003
LOG(PID)	-0.038146	0.016403	-2.325596	0.0301
C	0.349375	0.187105	1.867271	0.0759
R-squared	0.757998	Mean dependent var	-0.016300	
Adjusted R-squared	0.700378	S.D. dependent var	0.024345	
S.E. of regression	0.013326	Akaike info criterion	-5.605091	
Sum squared resid	0.003729	Schwarz criterion	-5.317127	
Log likelihood	81.66873	Hannan-Quinn criter.	-5.519464	
F-statistic	13.15522	Durbin-Watson stat	2.035487	
Prob(F-statistic)	0.000007			

*Note: p-values and any subsequent tests do not account for model selection.

Estimation Output of CA to GDP ratio Model

Table 8

Dependent Variable: LOG(GDP)
Method: ARDL
Date: 12/09/25 Time: 14:23
Sample (adjusted): 1999 2025
Included observations: 27 after adjustments
Maximum dependent lags: 1 (Automatic selection)
Model selection method: Akaike info criterion (AIC)
Dynamic regressors (0 lag, automatic): LOG(LBRFORCE) LOG(GFCF)
LOG(GOVTEXP)
Fixed regressors: C

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
LOG(GDP(-1))	0.169365	0.182554	0.927755	0.3636
LOG(LBRFORCE)	3.346450	0.909514	3.679383	0.0013
LOG(GFCF)	0.614626	0.247283	2.485520	0.0210
LOG(GOVTEXP)	-0.285625	0.119454	-2.391083	0.0258
C	-44.41437	12.62075	-3.519154	0.0019
R-squared	0.979409	Mean dependent var	12.27442	
Adjusted R-squared	0.975665	S.D. dependent var	0.513695	
S.E. of regression	0.080135	Akaike info criterion	-2.044626	
Sum squared resid	0.141276	Schwarz criterion	-1.804657	
Log likelihood	32.60245	Hannan-Quinn criter.	-1.973271	
F-statistic	261.6020	Durbin-Watson stat	2.140019	
Prob(F-statistic)	0.000000			

*Note: p-values and any subsequent tests do not account for model selection.

Estimation Output of GDP model

Table 9

Appendix B

Forecast: GDPF	
Actual: GDP	
Forecast sample: 1998 2064	
Adjusted sample: 1999 2064	
Included observations: 66	
Root Mean Squared Error	18029.39
Mean Absolute Error	14026.66
Mean Abs. Percent Error	5.817438
Theil Inequality Coef.	0.034717
Bias Proportion	0.000167
Variance Proportion	0.001809
Covariance Proportion	0.998024
Theil U2 Coefficient	0.504844
Symmetric MAPE	5.785843

CAGDPF results and coefficients

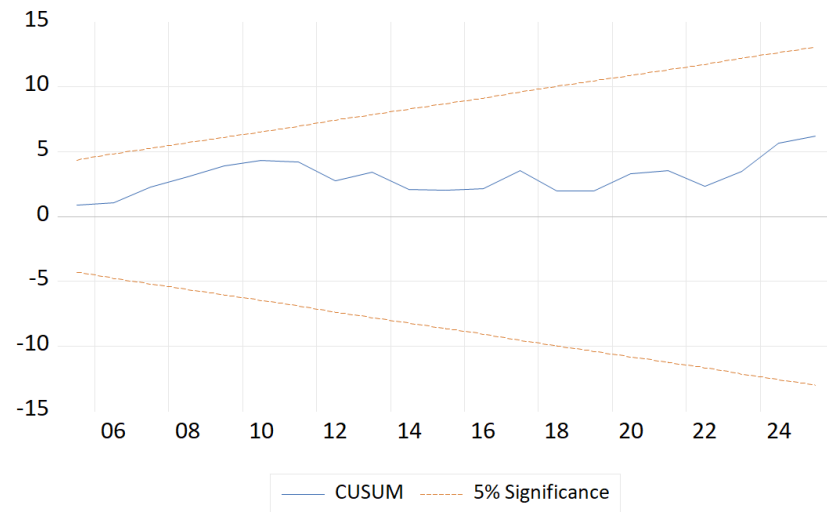
Figure 1

Forecast: CAGDPF	
Actual: CAGDP	
Forecast sample: 1998 2064	
Adjusted sample: 1999 2064	
Included observations: 66	
Root Mean Squared Error	0.013006
Mean Absolute Error	0.010517
Mean Abs. Percent Error	151.8628
Theil Inequality Coef.	0.243152
Bias Proportion	0.003141
Variance Proportion	0.141176
Covariance Proportion	0.855682
Theil U2 Coefficient	0.363034
Symmetric MAPE	78.25015

GDPF Results and coefficients

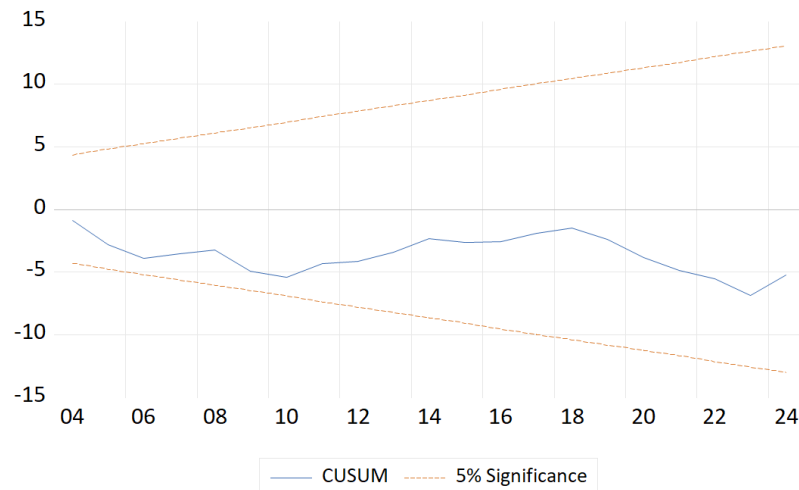
Figure 2

Appendix C



CUSUM CAGDP MODEL TEST

Figure 3



CUSUM GDP MODEL TEST

Figure 4